

B.A./B.Sc. (Hons.) SEMESTER V EXAMINATION 2022-23
STATISTICS

STA/STB-505: Practical based on Course No. STB-501 and STB-502

Time: Three hours

Max. Marks: 70

Note: The question paper contains three questions of equal marks. Attempt any **TWO** questions. Computation sheets and simulated values must be explicitly recorded.

1. (a) The following data relate to the area irrigated by the Damodar Canal in West Bengal over the years 1940-41 to 1948-49. Compute and describe the trend values of the series by applying: [cite: 4425, 4426]
- The method of moving averages (using a 3-year period)
 - The method of group averages
 - The method of semi-averages

Year	Area (in acres)	Year	Area (in acres)
1940-41	147,588	1945-46	171,569
1941-42	147,705	1946-47	168,989
1942-43	161,570	1947-48	181,888
1943-44	160,410	1948-49	178,868
1944-45	165,857		

[?]

- (b) Write a complete structured program in C to simulate a random sample of size n from a uniform population distributed over the interval $(0, \theta = 4)$. Use this logic to compute the empirical mean squared error (MSE) of the Maximum Likelihood Estimator (MLE) of θ . Report the explicit trend of the simulated MSE as the sample size increases across $n = 10, 20, 30, 50, 100$. [cite: 4429, 4430, 4431]
2. (a) Design an empirical test for the hypothesis $H_0 : \theta = 4.5$ against $H_1 : \theta = 9.5$ for a normal distribution with mean 0 and variance unity. Utilize a simulated sample of size $n = 25$ under the assumption that the true parameter $\theta = 4.5$, and trace the accompanying power-curve. [cite: 4432, 4433]
- (b) An incomplete segment of a life table is provided below. Compute all missing mathematical entries for the columns $(l_x, d_x, q_x, L_x, T_x, e_x^\circ)$, and estimate the conditional probability that a child of age 10 will live for at least 5 more years. [cite: 4434, 4435]

Age (x)	l_x	d_x	q_x	L_x	T_x	e_x°
10	100,000		0.00350			
11			0.00338			
12			0.00361			
13			0.00420			
14			0.00517			
15			0.00530			
16			0.00538			
17			0.00544			
18			0.00549			
19			0.00554		2,607,040	

[?]

3. (a) Simulate a random sample of size $n = 25$ from an exponential distribution with scale parameter $\theta = 2$ using MS EXCEL. Formulate and construct the 95% confidence intervals for θ using both the pivot method and large-sample structural approximations. On the basis of this simulated sample, test the hypothesis $H_0 : \theta = 2$ against $H_1 : \theta > 2$ and construct the matching power curve. [cite: 4440, 4441, 4442]
- (b) The table below records the wholesale prices and quantities of six primary agricultural crops during the years 2012 and 2015. Calculate Laspeyres's, Paasche's, and Fisher's ideal index numbers for the year 2015, using the year 2012 as the base. Provide an economic interpretation of your results. [cite: 4443, 4444]

	2012		2015	
Crop	Price (p_0)	Quantity (q_0)	Price (p_1)	Quantity (q_1)
Rice	207.92	1.50	306.67	6.25
Wheat	106.25	95.00	186.58	116.90
Jowar	1151.00	14.20	182.57	5.57
Barley	221.83	12.40	191.25	1.02
Bajra	256.75	13.15	155.75	6.15
Gram	203.00	1.05	497.83	0.61

[?]

External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>